Data Scientist Interview Guide

Original practice prompts and concise answer frameworks

A role-focused guide across analytics, statistics, experimentation, machine learning, and communication. Designed for retrieval by a resume-based interview preparation app.

How to use this guide

Treat each item as a practice prompt, not a script. Answer aloud, anchor claims in your own resume, and adapt the depth to the role and seniority. The sample answers are concise frameworks; add your own evidence and tradeoffs.

1. How do you approach an unfamiliar dataset?

Answer framework Clarify the decision the analysis must support, then inspect schema, grain, time coverage, missingness, duplicates, and quality constraints. Check leakage and sampling bias before summarizing distributions. Build a reproducible cleaning and analysis pipeline, validate key findings with slices or external context, and communicate uncertainty and the decision implication.

2. What is the difference between a confidence interval and a p-value?

Answer framework A confidence interval gives a range of parameter values compatible with the data under the procedure; a 95% procedure covers the true value in 95% of repeated samples. A p-value is the probability, assuming the null and model assumptions, of seeing results at least as extreme as observed. Neither is the probability the null is true; report effect size and practical relevance too.

3. How would you design an A/B test?

Answer framework State the hypothesis and primary outcome, define eligible units and randomization, choose guardrail metrics, estimate sample size from baseline rate, minimum detectable effect, alpha, and power, then run long enough to cover relevant cycles. Check assignment integrity and instrumentation before interpreting. Analyze by assigned group, report effect and uncertainty, and account for multiple tests or sequential monitoring.

4. A key product metric fell. How do you investigate?

Answer framework Verify the metric definition and data pipeline first. Decompose the change by funnel stage, cohort, platform, geography, and acquisition channel; compare numerator and denominator separately. Look for concurrent launches, seasonality, and mix shifts. Form testable causes, quantify contribution, and recommend a follow-up experiment or operational action.

5. When should you use precision versus recall?

Answer framework Precision matters when false positives are costly; recall matters when missing a true case is costly. For example, review queues may prioritize precision when capacity is tight, while safety screening often emphasizes recall. Choose a threshold against real costs and constraints, and report the confusion matrix and class prevalence alongside F1 or PR-AUC.

6. Explain the bias-variance tradeoff.

Answer framework A model with high bias cannot represent the pattern well and tends to underfit; high variance is too sensitive to the training sample and tends to overfit. Compare training and validation performance across appropriate splits. Reduce bias with richer features or model capacity; reduce variance with regularization, more representative data, simpler models, or ensembling.

7. How do you prevent leakage in model evaluation?

Answer framework Define the prediction time and only allow features available then. Split by time, entity, or group when the deployment setting requires it; fit imputers, encoders, and feature selection inside each training fold. Inspect target-derived fields and post-outcome events. Keep a final untouched test set and make the validation design mirror real use.

8. How do you choose a regression metric?

Answer framework Use MAE when typical absolute error and robustness to outliers matter; RMSE penalizes large misses more; MAPE can be unstable near zero and undefined at zero. Consider business costs, scale, and baseline comparisons. Always inspect residuals by segment and provide a simple benchmark so a metric score has context.

9. What does a p-value of 0.03 tell you?

Answer framework If the null hypothesis and test assumptions were true, data at least this incompatible with the null would occur about 3% of the time. It does not mean there is a 97% chance the effect is real. Check the effect estimate, interval, design quality, multiplicity, and whether the difference matters operationally.

10. How would you explain feature importance?

Answer framework Importance is model- and method-dependent, not proof of causality. Built-in tree gain can favor certain feature types; permutation importance measures score change after shuffling but can mislead with correlated predictors. Use holdout data, compare correlated features carefully, and explain whether the result is predictive or causal.

11. How do you decide if you have enough data?

Answer framework Learning curves show whether validation performance is still improving with more examples. Also assess uncertainty, subgroup coverage, label quality, rare-event counts, and stability across time. A large sample cannot repair systematic bias or leakage. Set an acceptable error or decision threshold and estimate the data needed to reach it.

12. What is the difference between correlation and causation?

Answer framework Correlation describes association; causation asks what would change under an intervention. Confounding, reverse causality, and selection can produce misleading associations. Randomization is a strong identification strategy; observational settings need a defensible design such as a natural experiment, adjustment with assumptions, or sensitivity analysis.

Research references

These references informed the topic selection and preparation emphasis. The prompts and answer frameworks in this PDF are newly written and paraphrased for study use; they are not reproductions of the linked articles.

- **GeeksforGeeks - AI ML DS Interview Series**
https://www.geeksforgeeks.org/interview-experiences/ai-ml-ds-interview/

- **GeeksforGeeks - Data Science Interview Questions and Answers**
https://www.geeksforgeeks.org/data-science/data-science-interview-questions-and-answers/

- **GeeksforGeeks - Lowe's India Data Scientist interview experience**
https://www.geeksforgeeks.org/interview-experiences/lowes-india-interview-experience-for-data-scientist-on-campus-2023/

- **Medium / Towards Data Science - Notes and technical questions from interviewing as a Data Scientist**
https://medium.com/towards-data-science/notes-and-technical-questions-from-interviewing-as-a-data-scientist-in-2018-20e7e3ee4ab3

- **Medium / Geek Culture - 70 Data Science Interview Questions**
https://medium.com/geekculture/70-data-science-interview-questions-you-need-to-know-before-your-next-technical-interview-ccfbd37a37b3

- **Microsoft Careers - Technical interviewing**
https://careers.microsoft.com/v2/global/en/hiring-tips/technical-interviewing

Knowledge-base note: this material is educational interview practice, not a guarantee of interview coverage. Tailor preparation to the role description, level, and candidate resume.